

Topic B18

Organisms and their environment

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B19.1	Energy flow
B19.2	Food chains and webs
B19.3	Nutrient cycles
B19.4	Population size
B19.5	Ecosystems

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Energy flow

The Sun is the main energy source. Producers transfer light energy into chemical energy; consumers obtain it by feeding. Energy is lost between trophic levels in respiration, movement, heat, waste and uneaten material, so food chains are short and biomass decreases upward.

Examiner insight

Energy flows and is eventually dissipated; nutrients cycle.

2. Food chains and pyramids

Arrows in a food chain show direction of energy transfer. Food webs reveal interdependence. Pyramids of biomass usually better represent available living material than numbers. Efficiency is energy transferred to the next level divided by energy available at the previous level.

Examiner insight

Do not draw arrows toward what is eaten; arrows point toward the consumer.

3. Carbon and nitrogen cycles

Photosynthesis removes carbon dioxide; feeding transfers carbon; respiration and decomposition return it; combustion rapidly releases stored carbon. Nitrogen-fixing bacteria convert nitrogen gas into compounds, nitrifying bacteria form nitrates, plants absorb nitrate, decomposers ammonify and denitrifying bacteria return nitrogen gas.

Examiner insight

Name the microorganisms or processes when explaining cycle arrows.

4. Populations and communities

Population size depends on births, deaths, immigration and emigration. Limiting factors include food, space, predation and disease. Sigmoid growth has lag, exponential and stationary phases; carrying capacity is the maximum sustainable population under current conditions.

Examiner insight

A stationary phase has continuing births and deaths at approximately equal rates.

Worked application

Situation: A top predator is removed from a food web.

Prediction: Its prey may increase and producers may decrease.

Reason: Reduced predation changes population balance and can create a trophic cascade.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B18

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Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines population? [1]
 - A all organisms of one species in an area at a given time
 - B all populations of different species in an area
 - C a community and its interactions with the physical environment
 - D the feeding position of an organism in a food chain
2. Which statement best defines community? [1]
 - A a community and its interactions with the physical environment
 - B all populations of different species in an area
 - C the feeding position of an organism in a food chain
 - D the maximum population an environment can sustain
3. Which statement best defines ecosystem? [1]
 - A the feeding position of an organism in a food chain
 - B the maximum population an environment can sustain
 - C a community and its interactions with the physical environment
 - D all organisms of one species in an area at a given time
4. Which statement best defines trophic level? [1]
 - A the maximum population an environment can sustain
 - B all organisms of one species in an area at a given time
 - C all populations of different species in an area
 - D the feeding position of an organism in a food chain
5. Which statement best defines carrying capacity? [1]
 - A the maximum population an environment can sustain
 - B all organisms of one species in an area at a given time
 - C all populations of different species in an area
 - D a community and its interactions with the physical environment
6. Which statement about energy flow is correct? [1]
 - A All species in an ecosystem.
 - B The Sun is the main energy source.
 - C One population and its habitat only.
 - D The non-living environment alone.
7. Which statement about food chains and pyramids is correct? [1]
 - A One population and its habitat only.
 - B The non-living environment alone.
 - C Arrows in a food chain show direction of energy transfer.
 - D The amount of water in a habitat.

8. Which statement about carbon and nitrogen cycles is correct? [1]
- A The non-living environment alone.
 - B The amount of water in a habitat.
 - C The fastest possible growth rate.
 - D Photosynthesis removes carbon dioxide; feeding transfers carbon; respiration and decomposition return it; combustion rapidly releases stored carbon.
9. Which statement about populations and communities is correct? [1]
- A Population size depends on births, deaths, immigration and emigration.
 - B The amount of water in a habitat.
 - C The fastest possible growth rate.
 - D All species in an ecosystem.
10. Which prediction is most appropriate when a top predator is removed from a food web? [1]
- A All species in an ecosystem.
 - B The non-living environment alone.
 - C Its prey may increase and producers may decrease.
 - D The fastest possible growth rate.

Section B · Structured questions

11 Biologists use precise vocabulary when studying organisms and their environment.

(a) Define population.

[2]

(b) Define community.

[2]

(c) Distinguish ecosystem from the incorrect idea: “the non-living environment alone”.

[2]

(d) State two important features of energy flow.

[2]

[Total: 8]

12 A top predator is removed from a food web.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records biomass in arbitrary units under three conditions.

condition	biomass in arbitrary units
producers	1200
primary consumers	180
secondary consumers	24

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to estimate plant population using quadrats.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of organisms and their environment to an unfamiliar context.

(a) Define trophic level. [2]

(b) Explain the biological importance of carbon and nitrogen cycles. [3]

(c) Correct this statement: “the fastest possible growth rate”. [2]

(d) State one further question that could be investigated. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. All organisms of one species in an area at a given time.
- 2 B. All populations of different species in an area.
- 3 C. A community and its interactions with the physical environment.
- 4 D. The feeding position of an organism in a food chain.
- 5 A. The maximum population an environment can sustain.
- 6 B. Energy flows and is eventually dissipated; nutrients cycle.
- 7 C. Do not draw arrows toward what is eaten; arrows point toward the consumer.
- 8 D. Name the microorganisms or processes when explaining cycle arrows.
- 9 A. A stationary phase has continuing births and deaths at approximately equal rates.
- 10 C. Reduced predation changes population balance and can create a trophic cascade.

Solutions · Structured questions

11. (a) all organisms of one species in an area at a given time [2]. (b) all populations of different species in an area [2]. (c) a community and its interactions with the physical environment; not the non-living environment alone [2]. (d) Credit any two accurate features from the guide: The Sun is the main energy source. Producers transfer light energy into chemical energy; consumers obtain it by feeding. Energy is lost between trophic levels in respiration, movement, heat, waste and uneaten material, so food chains are short and biomass decreases upward. [2].

12. (a) Its prey may increase and producers may decrease [2]. (b) Reduced predation changes population balance and can create a trophic cascade [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $1200 - 24 = 1176$ [2]. (b) The values change from 1200 to 24, with the highest value 1200. [2]. (c) Energy losses through respiration and waste reduce biomass at successive trophic levels [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Quadrat location [1]. (b) Organism count or percentage cover [1]. (c) Same quadrat size, sampling method, season and habitat [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Place many quadrats randomly and calculate a mean [1].

15. (a) the feeding position of an organism in a food chain [2]. (b) Award linked points from: Photosynthesis removes carbon dioxide; feeding transfers carbon; respiration and decomposition return it; combustion rapidly releases stored carbon. Nitrogen-fixing bacteria convert nitrogen gas into compounds, nitrifying bacteria form nitrates, plants absorb nitrate, decomposers ammonify and denitrifying bacteria return nitrogen gas. [3]. (c) Correct statement: carrying capacity means the maximum population an environment can sustain [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.